

PM Controller

Phase 0 — Action plan, milestones and build package · **tablet first**

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Client iPad first · SwiftUI · iOS 18+ **Backend** Supabase Postgres **Issued** 20 September 2026

1. What changed, and what did not

The original package specified a React mobile-first PWA. The client is now a **native iPad app**, with the phone layout following from the same binary. Everything underneath is unchanged — the schema, the row-level security, and all nine non-negotiable principles never depended on the client.

REQUIREMENT FROM THE PHASE 0 RISK LIST	PWA ON IPAD / IPHONE	NATIVE
Rework notification must reach the Lead with the app closed	Fragile — needs Home Screen install first	APNs. Reliable.
Close-out under 60 seconds, camera one tap	File-picker detour	Direct camera
A close-out with no signal must not be lost	Browser storage evicted under pressure	SwiftData, survives
Crews shoot 12 MP images all day	JavaScript-side compression	Hardware pipeline

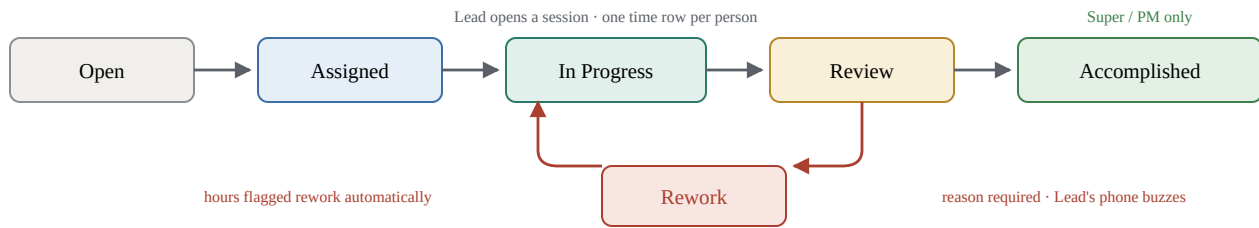
The cost, stated plainly. No Android, and App Store review on every release. Distribution to the Leads during the pilot is TestFlight, not the public store — faster, but capped at 100 internal testers with each build expiring after 90 days.

What it does *not* cost — seeing it work before paying for anything

WHAT YOU WANT TO DO	WHAT IT ACTUALLY COSTS
Compile it and run it in a Simulator	A Mac. No Apple account at all. ✓ available
Install it on an iPad you own	A Mac and a free Apple ID. Re-sign every 7 days. ✓ available
Put it on someone else's iPad (TestFlight)	\$99/yr Apple Developer Program
App Store	\$99/yr

A MacBook Pro is available, so the first two rows are already open: build it, run it in the iPad Simulator, and install it on your own iPad with a free Apple ID. The \$99 is only needed to put a build on *someone else's* device. CI still compiles on a clean machine and photographs every screen each run, which catches anything that builds locally only because of something already installed on that Mac.

2. Status flow — unchanged, and locked in the database



Six core statuses, enforced by a Postgres enum. A seventh cannot be inserted from any client. Custom board columns sit on top and each maps to exactly one of these, so reporting reads core status and never a column name.

3. Continuous integration

Three jobs on every pull request. The first is the one that matters most: SPEC principle 8 says access control lives in the database, and this is what stops that claim quietly decaying as later phases add tables.

JOB	RUNNER	WHAT IT PROVES
Row-level security	ubuntu	Both migrations applied to a real Postgres 16, then all 56 assertions. Non-zero exit on regression.
Build for iPad	macOS	The Swift compiles. Nobody needs a Mac for this to be true.
Screenshots on iPad	macOS	The app runs, photographed as each role, downloadable from the run.

4. Phase plan and milestones

#	DELIVERABLE	MILESTONE — CHECKED WHEN	STATE
0	Plan, model, flow, checklist	This package issued	DONE
1	Schema · auth · admin user setup · row-level security	Admin creates a Lead by email; that Lead sees only his project and cannot read the receipts tables at all	DATABASE GATE MET 56/56 assertions green. iOS shell awaiting a Mac build.
2	Task board, columns, role-gated transitions	A Lead cannot move a task to Accomplished	NEXT
3	Time capture — crew selection, per-person rows	One session with 3 crew writes 3 rows	—
4	Photos, description, quantity at close-out	Task closes with a photo from a phone in the field	—
5	Review gate · Rework column · push	Rejected task lands in Rework and the Lead's phone buzzes	—
6a	Cost class, change orders, custom columns	CO hours excluded from base earned value	—
6b	Schedule fields, GC baselines, blocked periods	Blocked days attribute to a named party	—

6c	Work class, discovered defects, source links	A defect ticket links back to an Accomplished task	—
6d	Material page — upload, extraction, categories	A photographed receipt lands categorised, duplicates caught	—
7	Progress page, every number drillable	Clicking any figure returns its ticket list; numbers reconcile to a hand-check of one week	—
8	Pilot on a fresh PCM job	Two full weeks run without paper backup	—
9	Closeout — results, improvements, business case	Final PDF issued	—

5. Phase 1 — what is built and proven

The Phase 1 milestone is a database claim, and it is now tested rather than asserted. Fifty-six assertions run as the role a phone actually holds — `authenticated` , with a real user id — not as the database owner, who bypasses every policy. The suite exits non-zero when a rule regresses, so it gates CI.

The matrix, proven

AREA	TESTS	WHAT IS ACTUALLY ASSERTED
Project isolation	9	A Lead sees exactly one project. A sub's Lead on another job sees none of the first job's tasks. A Manager sees every project. The same man is Foreman on one job and Super on another.
Material access	15	Lead reads zero receipts, zero line items, zero categories. A Foreman reads his own receipt and cannot reach the other Foreman's by id. PM, Assistant PM and Manager read everything.
Sign-off	4	Lead, Foreman and Assistant PM are all refused. Super is allowed.
History	7	An Accomplished task cannot be reopened, by anyone, including the Manager. <code>task_events</code> cannot be rewritten even by the service role.
Change orders	4	Assistant PM and Super refused in both directions. PM allowed.
Locked statuses	5	A seventh status is impossible. A column cannot map to a status it does not match. <code>rework_count</code> cannot be typed, only derived.
Lead movement	4	A Lead moves his own task and cannot touch another Lead's, create a task, or kick work back.
People & cost	8	A Lead cannot read <code>burdened_rate</code> at all. Nobody can be deleted — deactivation is the only path.

Six defects found and fixed in the reviewed schema

Four of them would have stopped the app working entirely. Full write-up in `docs/DECISIONS.md` .

REF	DEFECT	EFFECT IF SHIPPED
FIX-1	<code>citext</code> created after the table that uses it	Migration could not apply at all
FIX-3	<code>auth.uid()</code> is null for service-role writes, but <code>actor_id</code> is NOT NULL	Every seed and background status change fails
FIX-4	The rework counter trips the guard that protects its own column	No kick-back can ever be recorded
FIX-5	Cost view unusable as <code>security_invoker</code> under column grants	Nobody sees burdened rates, PM included

FIX-6	The audit trigger is blocked by its own table's RLS	No status change saves, for any role, with no error that points at the cause
FIX-2	<code>qty_needs_unit</code> ignored <code>qty_installed</code>	Close-out records "340" with no unit

6. Reminders & checklist

Before any more code — these block the build

- ☒ **Name the app and lock it.** PM Controller. Bundle identifier `com.apexmech.pmcontroller`.
- ☒ **Separate it from Apex.** An independent product; no contractor's name anywhere in the repo or the demo data.
- ☒ **A Mac is available.** MacBook Pro — building, Simulator and personal-device install are all open without an Apple Developer account.
- ☒ **Ownership settled.** Apex Plumbing and Mechanical Services SC owns it: LICENSE, a copyright header on every source file, and `NSHumanReadableCopyright` in the shipped bundle.
- ☐ **Confirm Apex controls `apexmech.com`.** The bundle identifier `com.apexmech.pmcontroller` assumes it. Cheap now, awkward after App Store submission.
- ☐ **Data-ownership terms with PCM in writing** before any of its data is entered. Apex owns the software; the customer owns its records — the LICENSE says so, a signed agreement settles it.
- ☐ **Create the Supabase project** (free tier is sufficient for the pilot) and apply `0001`, `0002`, then seed.
- ☐ **Confirm which iPads and iPhones the crews actually carry.** The floor is iOS/iPadOS 18 — anything that cannot reach 18 is deliberately out. Note there is no iOS 19–25: Apple went from 18 straight to 26.
- ☐ **Apple Developer Program enrolment** (\$99/yr) — only when you are ready to put builds on other people's devices. Not before.

Before the pilot goes live

- ☐ Written OK from PCM ownership before any PCM data lives in a system you own.
- ☐ Decide data ownership up front: who owns the records if the relationship ends.
- ☐ Decide burdened rates: real numbers, or placeholder multipliers until PCM shares them.
- ☐ Confirm the area / system list matches how PCM actually breaks out its estimates.
- ☐ Confirm the material category list against PCM's estimate breakdown.
- ☐ Agree the root-cause list for defects — short and fixed, never free text.
- ☐ Get the GC's published schedule in a usable form, or accept manual milestone entry per area.
- ☐ TestFlight build in every Lead's hands, with notifications actually tested on their phone.
- ☐ 15-minute walkthrough with each Lead — one screen, one rule: close the task before you leave.
- ☐ Run one week in parallel with existing daily reports and compare totals.
- ☐ CSV export path confirmed out of every table, so nothing is trapped.

Standing rules during the build

- ☒ Core statuses locked in the database, not hidden in the UI.

- ✓ Every status change writes to `task_events` , no exceptions.
- ✓ Never allow direct edit of hours; only start/end times and crew selection.
- ✓ Row-level security enforced in the database, proven by test, not by the UI.
- ✓ Metrics built as saved filters from the start, so drill-down is free rather than a rewrite.
- ✓ CI gates the access rules, so a later phase cannot quietly break them.
- ☐ Tablet layout first; the phone layout follows by size class from the same binary.
- ☐ Offline tolerance: a close-out with no signal must not be lost.
- ☐ Photo compression on upload — receipts must stay legible enough to audit against.
- ☐ Rework notification must reach the Lead with the app closed.

7. Risks still live

RISK	HANDLING
Leads don't close tasks in the field	Close-out under 60 seconds: crew already selected, camera one tap, quantity one field. Measured on a real phone in Phase 4, not assumed.
PCM says "that's just part of your job"	Apex owns PM Controller outright — asserted in LICENSE and in every source file. PCM is the pilot customer, not a co-owner. Get it in writing before go-live.
Rework becomes a blame tool	Reasons describe the condition, not the person. Defect rate is never shown per individual — by system, area and phase only. This is built in, not policy.
Scope creep into a full ERP	v1 ships with no scheduling, no invoicing, no payroll export, no Android.
App Store review delays a fix	TestFlight for the pilot; expedited review exists for genuine blockers. Budget 24–48 hours for any store release.
Data trust	Phase 7 is not done until a week of app numbers matches a hand tally.

Next artifact: Phase 1 completion PDF — full chat interactions plus summary — issued once the app has been built green in CI and a real Lead has signed in against a live Supabase project. The database half of Phase 1 is already proven and is recorded above.